EEE4114F Class Test 2

June 2021

Duration: 1 hours | Total: 40 marks

Empl ID: | | | | | | | |

Instructions:

- This is a closed-book test but you may use a calculator.
- There are two parts to this test.
- **Part A** has *two* questions totalling 20 marks. You must answer all of them.
- **Part B** has *two* questions totalling 20 marks. You must answer all of them.
- Part A and part B each count 50% of the final mark.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 1 hour.

Part A:

Question:	1	2	Total
Points:	10	10	20
Score:			

Part B:

Question:	3	4	Total
Points:	10	10	20
Score:			

Part A: Digital Signal Processing

Question 1 [10 marks]

The continuous-time signal

$$x_c(t) = \frac{\sin(4000\pi t)}{4000\pi t}$$

is sampled with a sampling period T to obtain the discrete-time signal

$$x[n] = \frac{\sin(\pi n/3)}{\pi n/3}.$$

a. Determine a choice for T consistent with this information.

b. Is your choice for T above unique? If so, explain why. If not, specify another choice of T consistent with the information given.

c. Specify and sketch $X(e^{j\omega})$ over the range $0 \leq \omega \leq 4\pi$.

Solution:

a. The sampling process leads to

$$x[n] = x_c(nT) = \frac{\sin(4000\pi Tn)}{4000\pi Tn} = \frac{\sin(\pi n/3)}{\pi n/3}.$$

Clearly this is true when $4000\pi T = \pi/3$, or $T = 1/12000$ seconds.

b. The value of T above is unique because $x_c(t)$ is not a periodic function.

c. From tables of transforms we know that

$$\frac{\sin(\omega_c n)}{\pi n} \overset{\mathcal{F}}{\longleftrightarrow} p_{2\omega_c}(\omega),$$

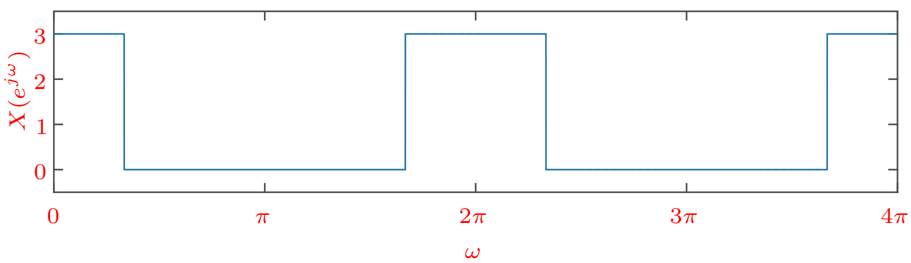
where $p_\tau(\omega)$ is a unit pulse of total width τ centred on the origin. Thus

$$\frac{\sin(\omega_c n)}{\omega_c n} \overset{\mathcal{F}}{\longleftrightarrow} \frac{\pi}{\omega_c} p_{2\omega_c}(\omega)$$

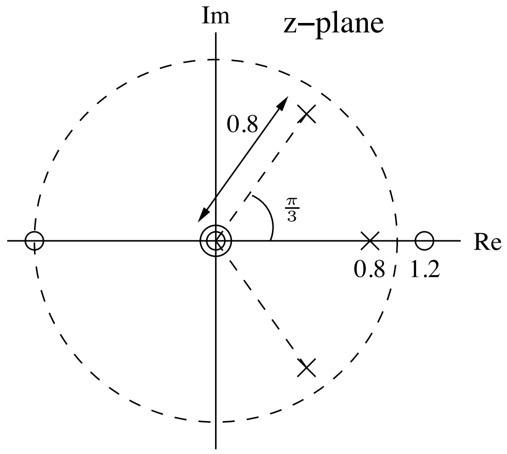
and setting $\omega_c = \pi/3$ gives

$$X(e^{j\omega}) = 3p_{2\pi/3}(\omega).$$

Sketch below:



Question 2 [10 marks]
Consider an LTI system with the following pole-zero configuration:



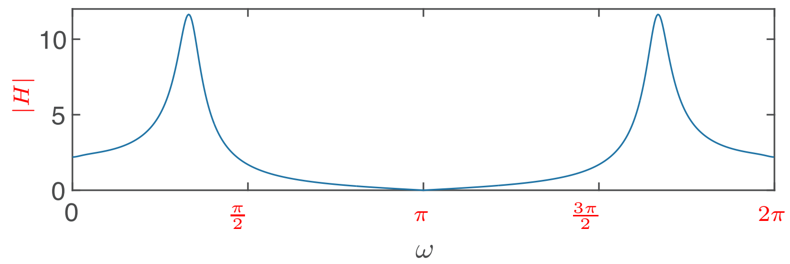
a. Give an expression for the system function $H(z)$.
b. Sketch the magnitude response of the system.
c. What type of filter does this system represent?
d. What is the approximate phase response of the system at $\omega = 0$?

Solution:

a. From the locations of the zeros and poles we can formulate the system function

$$H(z) = K\frac{z^2(z+1)(z-1.2)}{(z-0.8)(z-0.8e^{-\pi/3})(z-0.8e^{\pi/3})}$$

b. Product of lengths of zero vectors divided by product of length of pole vectors should give you something like



c. The filter has a bandpass characteristic, with the response peaking at around $\omega = \pi/3$ (passes low frequencies, amplifies frequencies around $\pm\pi/3$, and attenuates high frequencies).

d. At $\omega = 0$, sum of angles of zero vectors is $0 + 0 + 0 + \pi = \pi$, and sum of angles of pole vectors is approximately $(2\pi - \frac{\pi}{3}) + 0 + \frac{\pi}{3}) = 0$. The phase is therefore π (or $-\pi$).

END OF PART A

Part B: Machine Learning

Question 3 [10 marks]

Suppose you are developing a self-driving car that has access to a forward-facing camera.

You train three different models: A, B, and C, as part of a task to identify street signs from the camera. You train and evaluate the models using 5-fold cross-validation with results shown as a box plot in Figure 1.

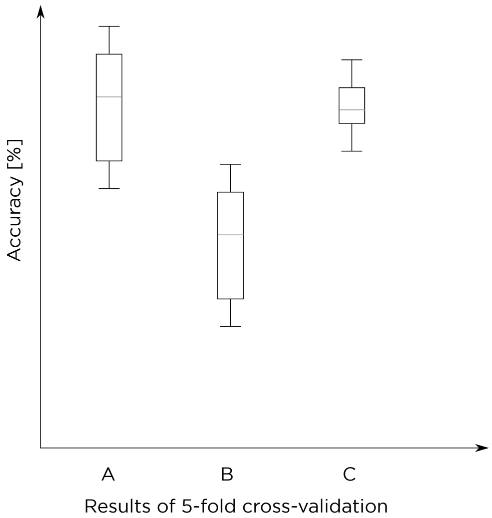


Figure 1: Box plots for three trained models using 5-fold cross-validation

a. Explain what k-fold cross-validation is and why it is useful for evaluating a model's ability to generalise to unseen data compared to training and evaluating with a single holdout split. (2)

Solution: k-fold cross-validation repeats the splitting process k times and ensures that the validation data on each split is independent. In this way there would be multiple validation results, each on completely different examples which provides a better estimate than if there was only a single validation/test set that may not contain a good representation of the entire data set.

b. Figure 2 shows a plot of model capacity vs error. Which points on this plot (points indicated as 1, 2, or 3 on the x-axis) would you expect to correspond to each model (A, B, and C)? Explain your decisions. (6)

Solution:

1 Model B, shows accuracy results in the box-and-whisker plot that indicate high bias, as the accuracy values are mostly lower than for models A and C. It shows high variance as well, based on the large spread of accuracy values when compared to model C. The model is underfit and likely has too low of a capacity for the problem.

2 Model C, accuracy values are high compared to model B, and almost comparable to model A which indicates a reasonable level of bias. The spread of accuracy values is low which is an indication of low variance. The model appears to be well fit to the problem and has enough capacity to learn but not so much that it overfits.

3 Model A, the model shows instances of very high accuracy, but is hampered by a large variance in the results. The model likely has a greater capacity which can lead to low training errors, but it can overfit to the training data, which results in high variance when evaluated on unseen data.

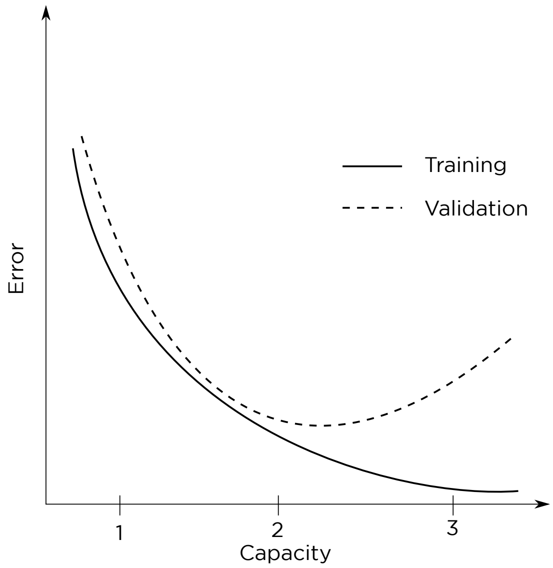


Figure 2: Model capacity vs. error

c. Which model would you choose if you wanted the real-world performance to be reliable and close to the predicted performance. Explain your answer. (2)

Solution: Model C, as it shows lower variance in the box-and-whisker plot compared to the other models. The performance estimates are more likely to generalise to the real-world.

Question 4 [10 marks]

Suppose you need to develop a system that measures distance to an obstacle and predicts a safe speed to travel at. You collect a data set, $\mathcal{D}$, from a professional driver consisting of input-output pairs (x_i, y_i) where x_i is the distance to an obstacle measured in m, and y_i is the target variable, which in this case is the speed of the vehicle measured in m/s.

$$\mathcal{D} = \{(1, 4.8), (3, 15.2), (1.5, 7.3), (2, 10.2)\}$$

You then decide to fit a linear regression model of the form $h_\theta(x) = \theta_0 + \theta_1 x$ to it using gradient descent optimization methods. The model's parameters are randomly initialised such that

$$\theta_0 = 0.1, \theta_1 = 0.3$$

. The objective function, J, you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{N}\sum_{i=1}^{N}(h_\theta(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

Suppose you decide to divide the data set into two, equally sized minibatches, keeping the order as it is such that

$$\mathcal{D} = \{((1, 4.8), (3, 15.2)), ((1.5, 7.3), (2, 10.2))\}$$

A single training iteration uses one minibatch and the model parameters are updated after **each** iteration. The update rule is given by:

$$\theta_i := \theta_i - \eta\frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.001$. Answer the following questions

a. Write down the expressions for the gradients of the objective function with respect to the model's parameters $\frac{\partial J}{\partial \theta_0}$ and $\frac{\partial J}{\partial \theta_1}$. (2)

Solution:

$$\frac{\partial J}{\partial \theta_0} = \frac{2}{2}\left(\sum_{i=1}^{2}((\theta_0 + \theta_1 x_i) - y_i)\right)$$
$$\frac{\partial J}{\partial \theta_1} = \frac{2}{2}\left(\sum_{i=1}^{2}((\theta_0 + \theta_1 x_i) - y_i)x_i\right)$$

b. Calculate the gradients of the parameters after one training iteration using the first minibatch. (2)

Solution:

$$h_\theta(1) = 0.1 + 0.3(1) = 0.4$$
$$h_\theta(3) = 0.1 + 0.3(3) = 1$$
$$\frac{\partial J}{\partial \theta_0} = ((0.4 - 4.8) + (1 - 15.2)) = -18.6$$
$$\frac{\partial J}{\partial \theta_1} = ((0.4 - 4.8)(1) + (1 - 15.2)(3)) = -47$$

c. Calculate the **model** parameters after the second training iteration using the second minibatch. (4)

Solution: First we need to know the parameters after the first update.

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta\frac{\partial J}{\partial \theta_0} \\ &:= 0.1 - 0.001(-18.6) \\ &:= 0.1186 \\ \theta_1 &:= \theta_1 - \eta\frac{\partial J}{\partial \theta_1} \\ &:= 0.3 - 0.001(-47) \\ &:= 0.3470\end{aligned}$$

Then repeat the calculation of gradients, but using the new parameters and new data $h_\theta(x_i) = 0.1186 + 0.3470(x_i)$

$$\begin{aligned}h_\theta(1.5) &= 0.1186 + 0.3470(1.5) = 0.6391 \\ h_\theta(2) &= 0.1186 + 0.3470(2) = 0.8126 \\ \frac{\partial J}{\partial \theta_0} &= ((0.6391 - 7.3) + (0.8126 - 10.2)) = -16.05 \\ \frac{\partial J}{\partial \theta_1} &= ((0.6391 - 7.3)(1.5) + (0.8126 - 10.2))(2) = -28.77\end{aligned}$$

Then the new parameters are

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta\frac{\partial J}{\partial \theta_0} \\ &:= 0.1186 - 0.001(-16.05) \\ &:= 0.135 \\ \theta_1 &:= \theta_1 - \eta\frac{\partial J}{\partial \theta_1} \\ &:= 0.3470 - 0.001(-28.77) \\ &:= 0.376\end{aligned}$$

d. Your friend tells you increase the capacity of you model by adding more input features including: the colour of the obstacle, the day of the week, and the ambient temperature. Would it make sense for this application to use l_1 or l_2 regularization terms? Explain your answer. (2)

Solution: It does not seem as though any of those features should have any impact on how fast your vehicle should approach the obstacle. It would make more sense to choose an l_1 regularization term to eliminate the useless features, as it would reduce the parameters to 0, whereas l_2 would keep them and just minimize their effect.

END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt \iff x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t}d\Omega \qquad \text{(CTFT)}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \iff x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} \qquad \text{(DTFT)}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \iff x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz \qquad \text{(ZT)}$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \iff x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} \qquad \text{(DFT)}$$

Discrete-time Fourier transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(e^{j\omega})$, $Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d}X(e^{j\omega})$	Time shift
$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$1 \quad (-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (\|a\| < 1)$	$\frac{1}{1-ae^{-j\omega}}$
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n + 1)a^n u[n] \quad (\|a\| < 1)$	$\frac{1}{(1-ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \|\omega\| < \omega_c \\ 0 & \omega_c < \|\omega\| \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)}e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(z)$, $Y(z)$	ROC	Property
$ax[n]+by[n]$	$aX(z)+bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n-n_d]$	$z^{-n_d}X(z)$	ROC $= R_x$	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC $= \lvert z_0\rvert R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC $= \frac{1}{R_x}$	Time reversal
$nx[n]$	$-z\frac{dX(z)}{dz}$	ROC $= R_x$	Frequency diff.
$x[n]*y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC $= R_x$	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
$\begin{cases} a^n & 0 \le n \le N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$\lvert z\rvert > 0$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$r^n\cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > \lvert r\rvert$

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for} \quad a \neq 1.$$